



Course: Electro Mechanical Systems
 Program: BS(Electrical Engineering)
 Duration: 60 Minutes
 Paper Date: 04-Nov-17
 Section: ALL
 Exam: Mid2 Examination

Course Code: EE221
 Semester: Fall 2017
 Total Marks: 60
 Weight: 15%
 Page(s): 6
 Questions: 3

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Section: EE-2

Instruction/Notes:

- Don't ask any questions; in case of an ambiguity, use valid assumptions if needed.
- State the units of each value where necessary.

Q1 A 2500-V, ^{VT}1200-kVA^S, 0.8-PF-lagging, 60-Hz, 4 pole, Y-connected synchronous generator, has a synchronous reactance of 1.2Ω and an armature resistance of 0.2Ω . [20] CLO
 At 60 Hz, its friction and windage losses are 30 kW, and its core losses are 20 kW. 04
 The field circuit has a dc voltage of 220 V, and the maximum I_F is 10 A. The resistance of the field circuit is adjustable over the range from 22 to 200 Ω .
 The field current has been adjusted to 5.5A and a delta connected load having per phase impedance of $18.186 + j10.5 \text{ ohm}$ is connected to the generator. OCC of the generator is given below.

Find out the following quantities.

- 1) voltage regulation
- 2) efficiency
- 3) output torque of prime mover

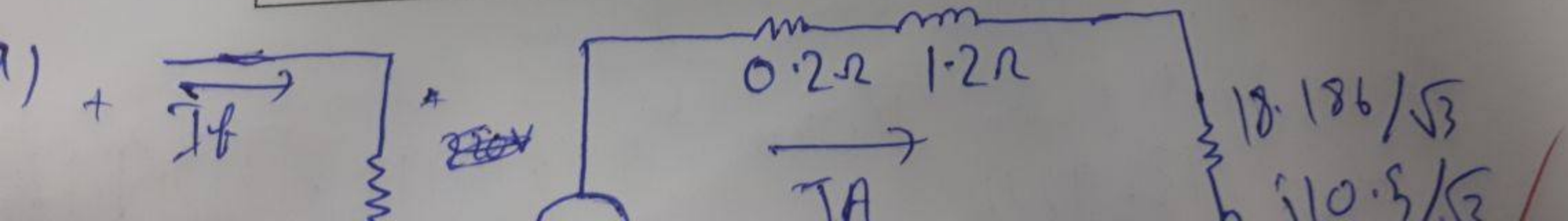
I_F (A)	2	4	6	8	10
$V_{T,NL}$ (V)	1200	2200	2600	2750	2850

OCC for above generator

$$VR = \frac{|E_{af}| - V_{\phi} \times 100}{V_{\phi}}$$

$$\frac{2850 - 2200}{2200} \times 100 = 30\%$$

$$5.5 - 4$$



National University of Computer and Emerging Sciences, Lahore Campus



Course: Electro Mechanical Systems
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Course Code: EE221
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Name: H. Yasir

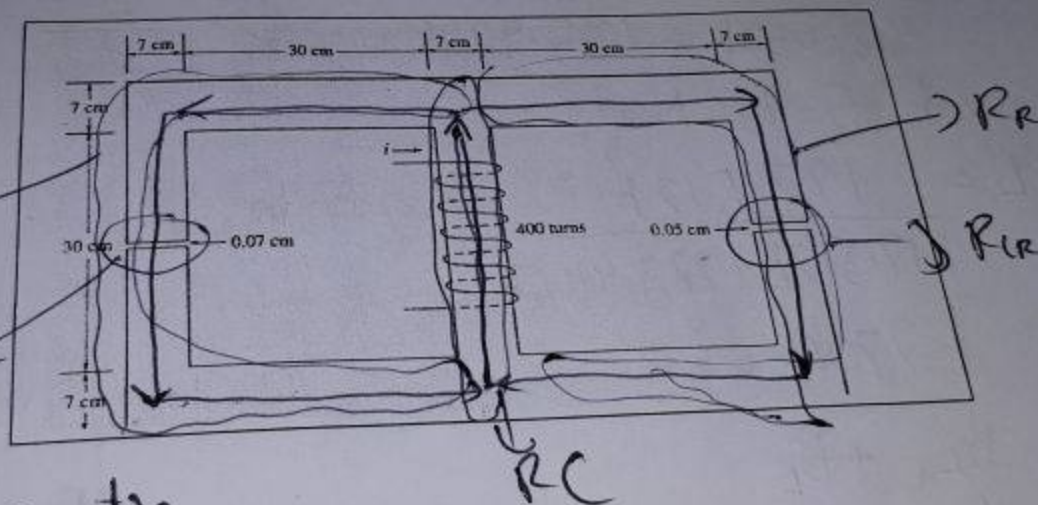
Roll No.: 13L-4307

Section: EE-3

Instruction/Notes:

- Use valid assumptions if needed.
- State the units of each value where necessary.

Q1 A ferromagnetic core with a relative permeability of 1500 is shown in Figure below. [20]
 The dimensions are as shown in the diagram, and the depth of the core is 7 cm. The air gaps on the left and right sides of the core are 0.070 and 0.050 cm, respectively. CLO
 Because of fringing effects, the effective area of the air gaps is 5 percent larger than 01
 their physical size. If there are 400 turns in the coil wrapped around the center leg of the core and if the current in the coil is 1.0 A, what is the flux in the left leg of the core as well as the flux density in right air gap?

~~Total RL~~

RL = Reluctance of left side

RR = reluctance of right side

RL = reluctance of air gap right

RL = reluctance of air gap left

RC = reluctance of centre

$$R_L = \frac{l_c}{\mu_0 \mu_r A}$$

$$= \frac{1.11}{1500 \times 4\pi \times 10^{-7} \times (0.07 \times 0.07)}$$

$$R_R = \frac{1.11}{1500 \times 4\pi \times 10^{-7} \times (0.07 \times 0.07)}$$

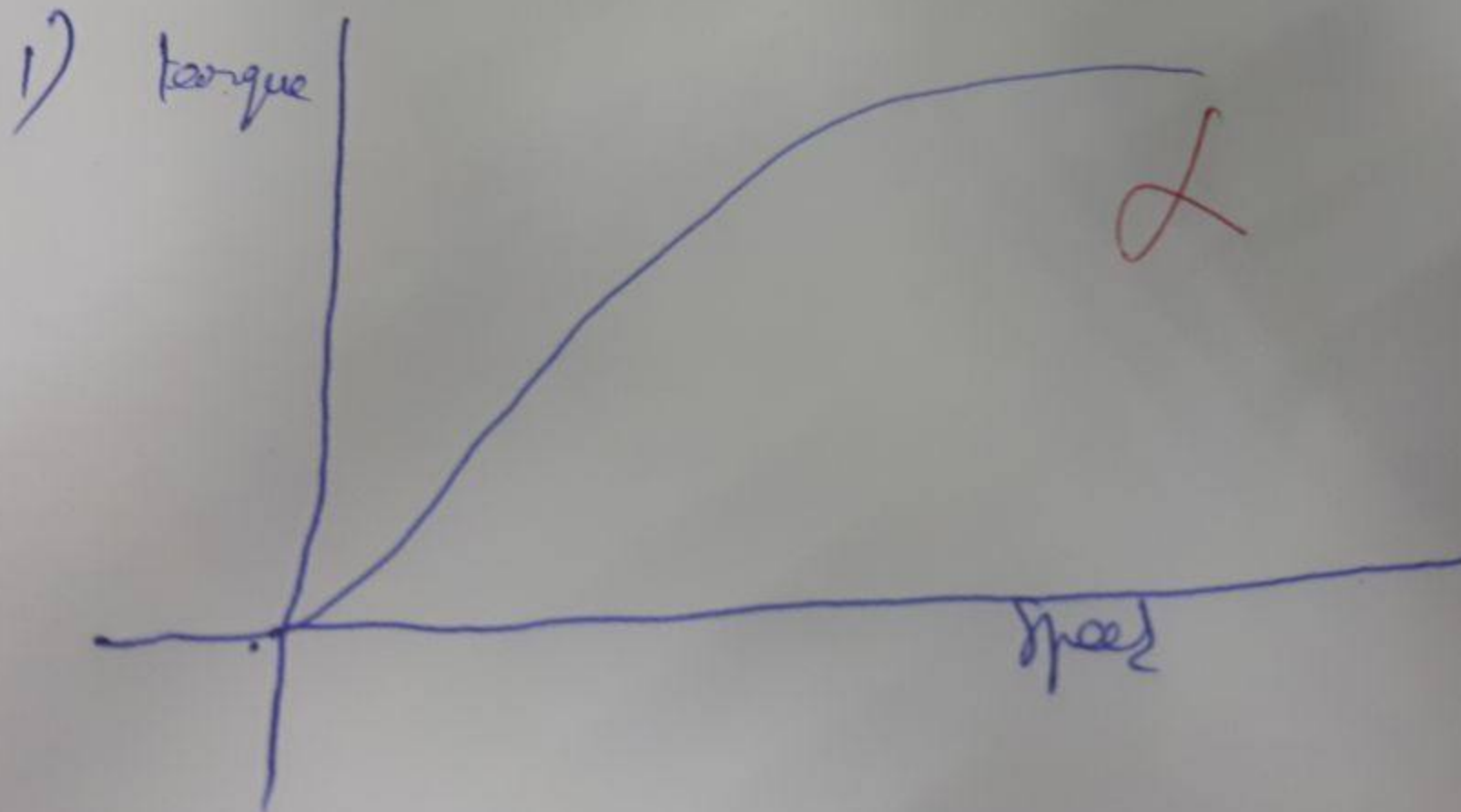
$$= 120.178 \text{ K} \frac{\text{at}}{\text{wb}}$$

$$R_C = \frac{0.31}{1500 \times 4\pi \times 10^{-7} \times (0.07 \times 0.07)}$$

$$= 40.059 \text{ K} \frac{\text{at}}{\text{wb}}$$

Q3 Objective type: short answers, diagrams etc [Binary Marking]

- 1) Draw the torque speed curve of a compensated and uncompensated DC shunt motor on the same coordinate axis [20] CLO
- 2) A 250 V 150A DC shunt motor has $R_A = 0.1 \text{ ohm}$ and $R_F = 250 \text{ ohm}$, it has a three stage starter circuit consisting of three equal resistances, what should be the value of each resistance if the starting current at the time of switch closing should not exceed double the full load current. 03 04
- 3) Draw the torque speed curve of a DC Series Motor
- 4) Draw the phasor diagram of an alternator to show the effect of change of inductive load keeping I_F constant (ignore R_A) (begin)
- 5) A delta connected synchronous generator was tested for a DC test. When a DC voltage of 10 V was applied to two of the terminals, a current of 30 A was measured. What is the value of per phase armature resistance.



- Q2 A fully compensated 250-V 110-A dc Shunt motor has following parameters
 $R_A = 0.15 \Omega$ $N_F = 1500$ turns $N_A = 80$ turns
 $R_F = 200 \Omega$ $n_m = 1250$ r/min
 $R_{adj} = 10$ to 300Ω
 Find out following

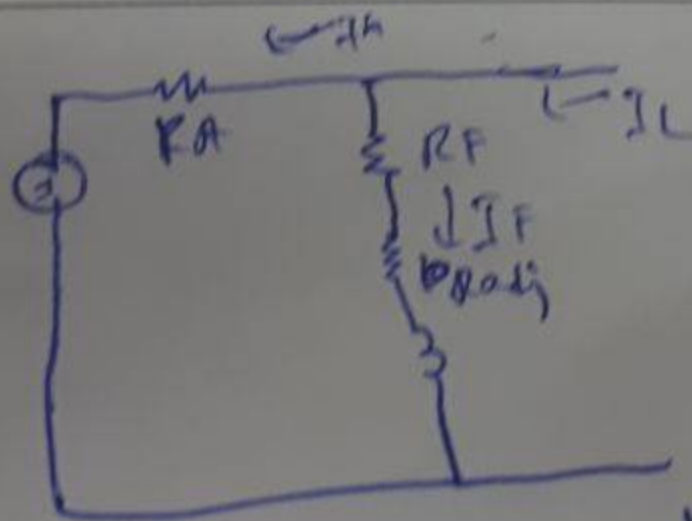
- (a) Under no-load conditions, what range of possible speeds can be achieved by adjusting R_{adj}
 (b) What is full-load speed of this motor when $R_{adj} = 100 \Omega$?
 (c) What is induced torque in part (b)

This motor has compensating windings and interpoles. The magnetization curve for this motor is shown in Figure below

Magnetization Curve for above motor at speed = 1200 rpm

I_f (A)	0.25	0.5	0.75	1	1.25
E_A (V)	100	200	270	285	290

(a)



$E_A = V_T$ at no load
 $I_L = I_A + I_F$

$V_T = E_A + I_A R_A$

for $R_{adj} = 10 \Omega$

at no load, $E_A = 250V$
 I_F

$\frac{270 - 200}{0.75 - 0.5} = 25$

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Page(s): 3
Questions: 3

Instruction/Notes:

- Do not ask any questions; write down your valid assumptions if required.
- State the **units** of each value clearly.

Q1 A 2.5KV, 1400-kVA 50-Hz eight-pole Y-connected synchronous generator has a synchronous reactance of 1.2Ω and an armature resistance of 0.18Ω . Its friction and windage and core losses are 65 kW. The generator feeds a balanced Y connected load having per phase impedance of $3.333 \angle 30^\circ \Omega$. A line voltage of 2078.5 V appears across load terminals. Find out the following quantities under these conditions.

- 1) The line current
- 2) Voltage regulation
- 3) Efficiency
- 4) Applied Torque by the Prime Mover